

XML Namespaces

- XML Namespaces provide a method to avoid element name conflicts.

XML document_1

```
<table>  
  <tr>  
    <td>Apples</td>  
    <td>Bananas</td>  
  </tr>  
</table>
```

XML document_2

```
<table>  
  <name>African Coffee Table</name>  
  <width>80</width>  
  <length>120</length>  
</table>
```

- Solution : Adding prefixes

XML document_1

```
<h:table>  
  <h:tr>  
    <h:td>Apples</h:td>  
    <h:td>Bananas</h:td>  
  </h:tr>  
</h:table>
```

XML document_2

```
<f:table>  
  <f:name>African Coffee Table</f:name>  
  <f:width>80</f:width>  
  <f:length>120</f:length>  
</f:table>
```

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  <tr>
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  </tr>
</table>
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XML document_2

```
<table>
  <name>African Coffee Table</name>
  <width>80</width>
  <length>120</length>
</table>
```

- Defining a namespace for each prefix: **use an xmlns attribute**

XML document_1

```
<h:table xmlns:h="http://www.w3.org/TR/html4/">
  <h:tr>
    <h:td>Apples</h:td>
    <h:td>Bananas</h:td>
  </h:tr>
</h:table>
```

XML document_2

```
<f:table xmlns:f="https://www.w3schools.com/furniture">
  <f:name>African Coffee Table</f:name>
  <f:width>80</f:width>
  <f:length>120</f:length>
</f:table>
```

The namespace can be defined by an xmlns attribute in the start tag of an element.

XML Namespaces

- XML Namespaces provide a method to avoid element name conflicts.
- The namespace can be defined by an xmlns attribute in the start tag of an element.

XML document_1

```
<table>  
  <tr>  
    <td>Apples</td>  
    <td>Bananas</td>  
  </tr>  
</table>
```

XML document_2

```
<table>  
  <name>African Coffee Table</name>  
  <width>80</width>  
  <length>120</length>  
</table>
```

- Defining the default namespaces

XML document_1

```
<table xmlns="http://www.w3.org/TR/html4/">  
  <tr>  
    <td>Apples</td>  
    <td>Bananas</td>  
  </tr>  
</table>
```

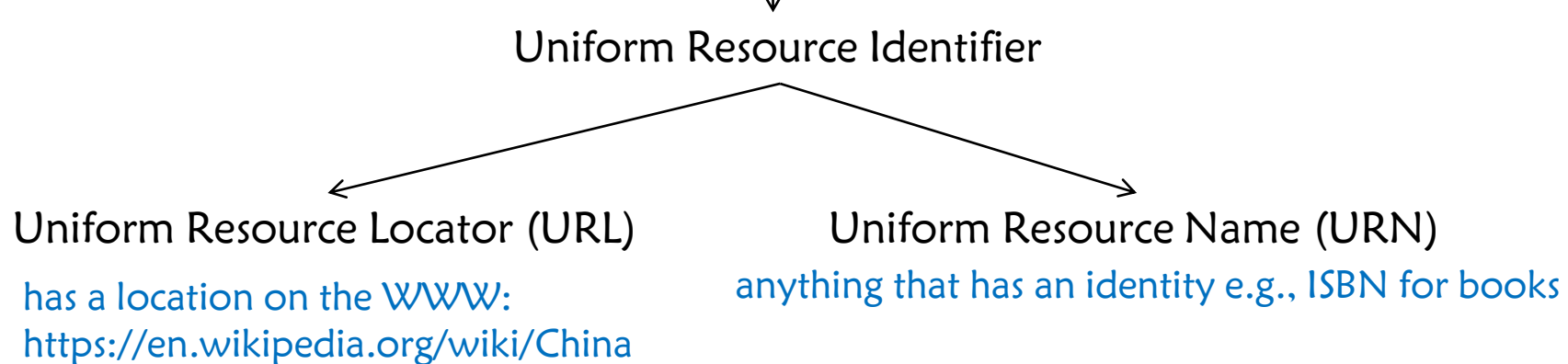
XML document_2

```
<table xmlns="https://www.w3schools.com/furniture">  
  <name>African Coffee Table</name>  
  <width>80</width>  
  <length>120</length>  
</table>
```

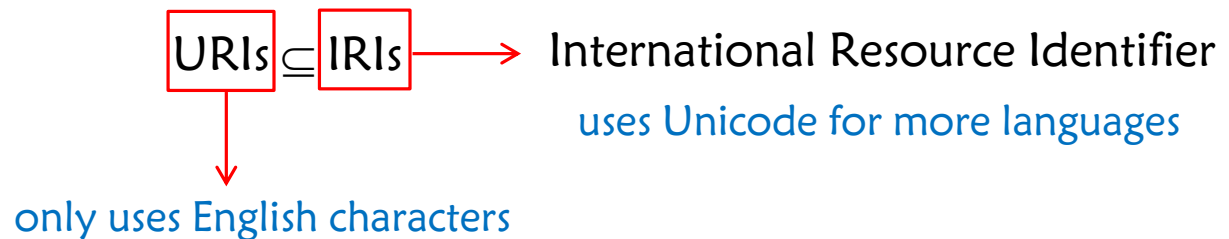
XML Namespaces

- XML Namespace Syntax:

`xmlns: namespace-prefix= "namespaceURI"`



$\text{URLs} \subseteq \text{URIs}$



XML Namespaces

- URI (IRI) format

URI = scheme:[//authority]path[?query][#fragment]

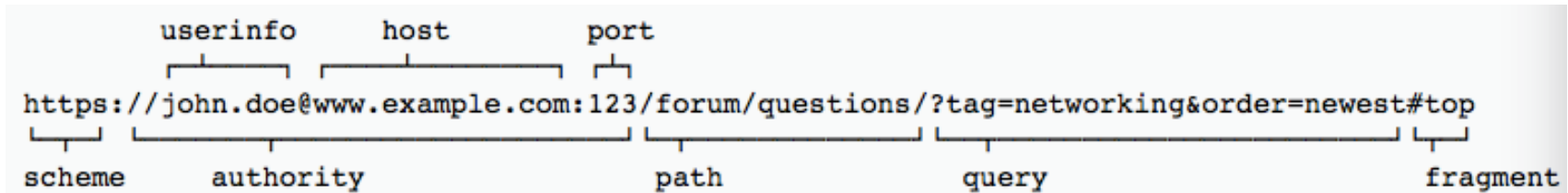
-scheme: type of URI, e.g. http, ftp, mailto, file, urn

-authority: optional; typically a domain name

-path: e.g., file system path

-query: optional; provides non-hierarchical information. Usually for parameters, e.g. for a web service

-fragment: optional; often used to address part of a retrieved resource, e.g. section of a HTML file; the set of characters is limited to US-ASCII



urn:isbn:0451450523

↓
scheme

↓
path